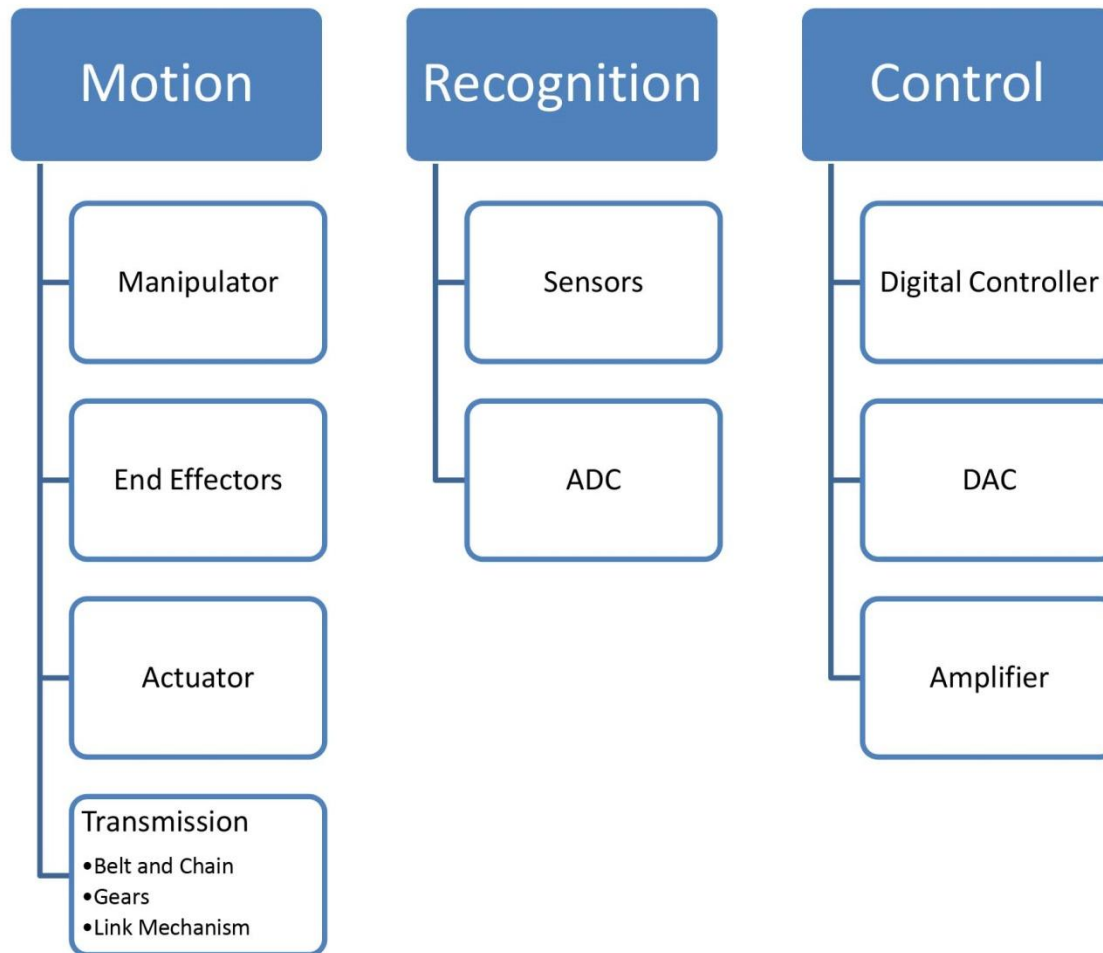


Introduction to Robotics

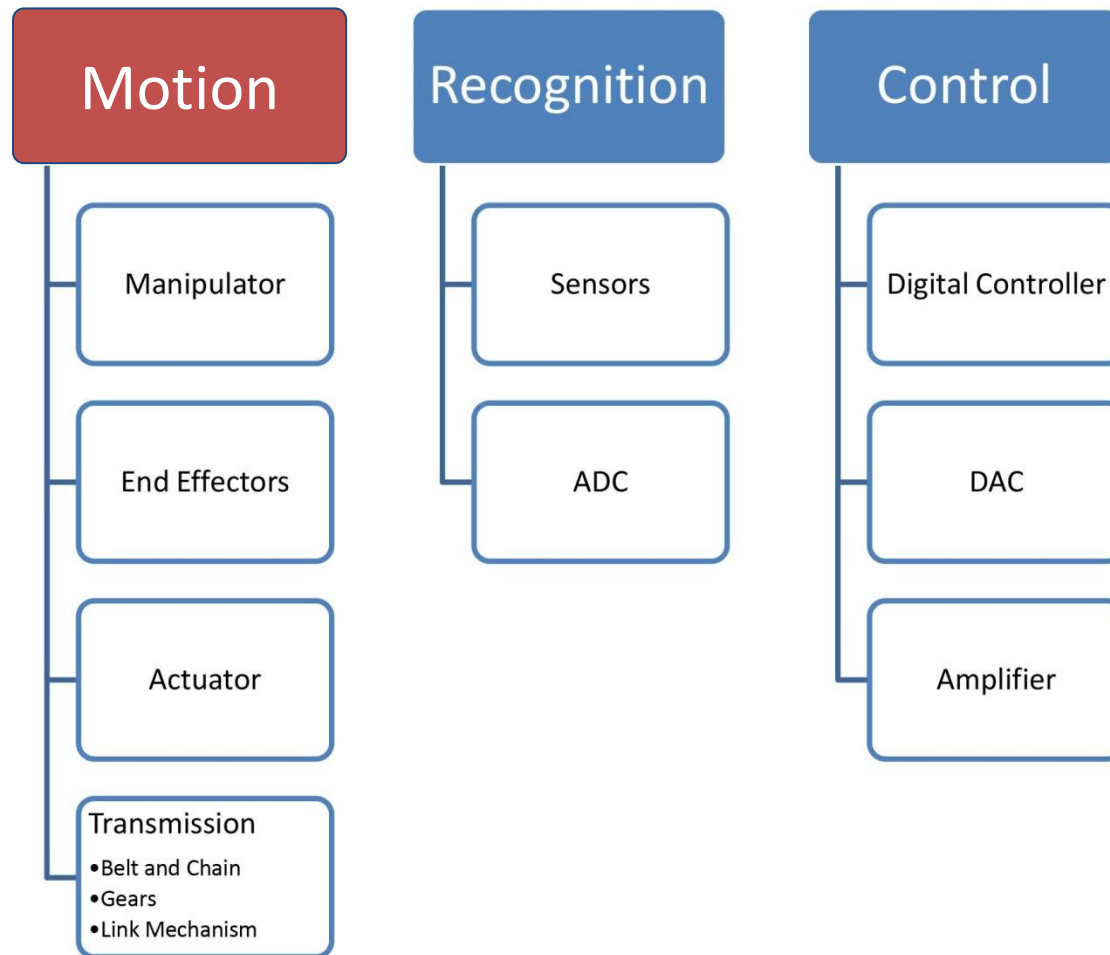
CSE461

BRAC University

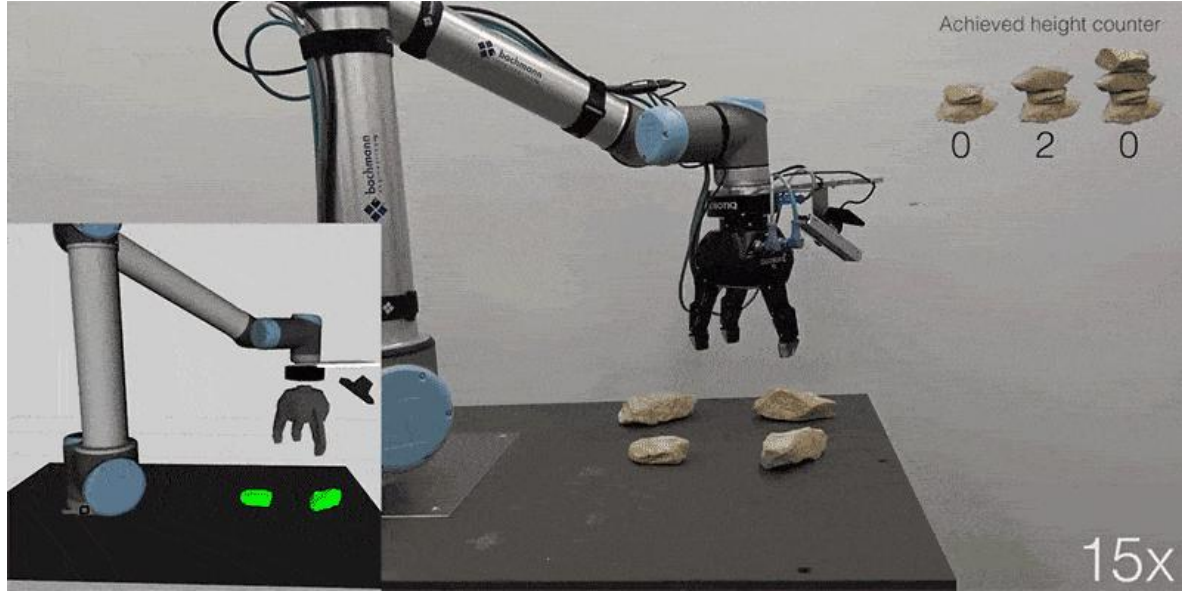
Subsystems



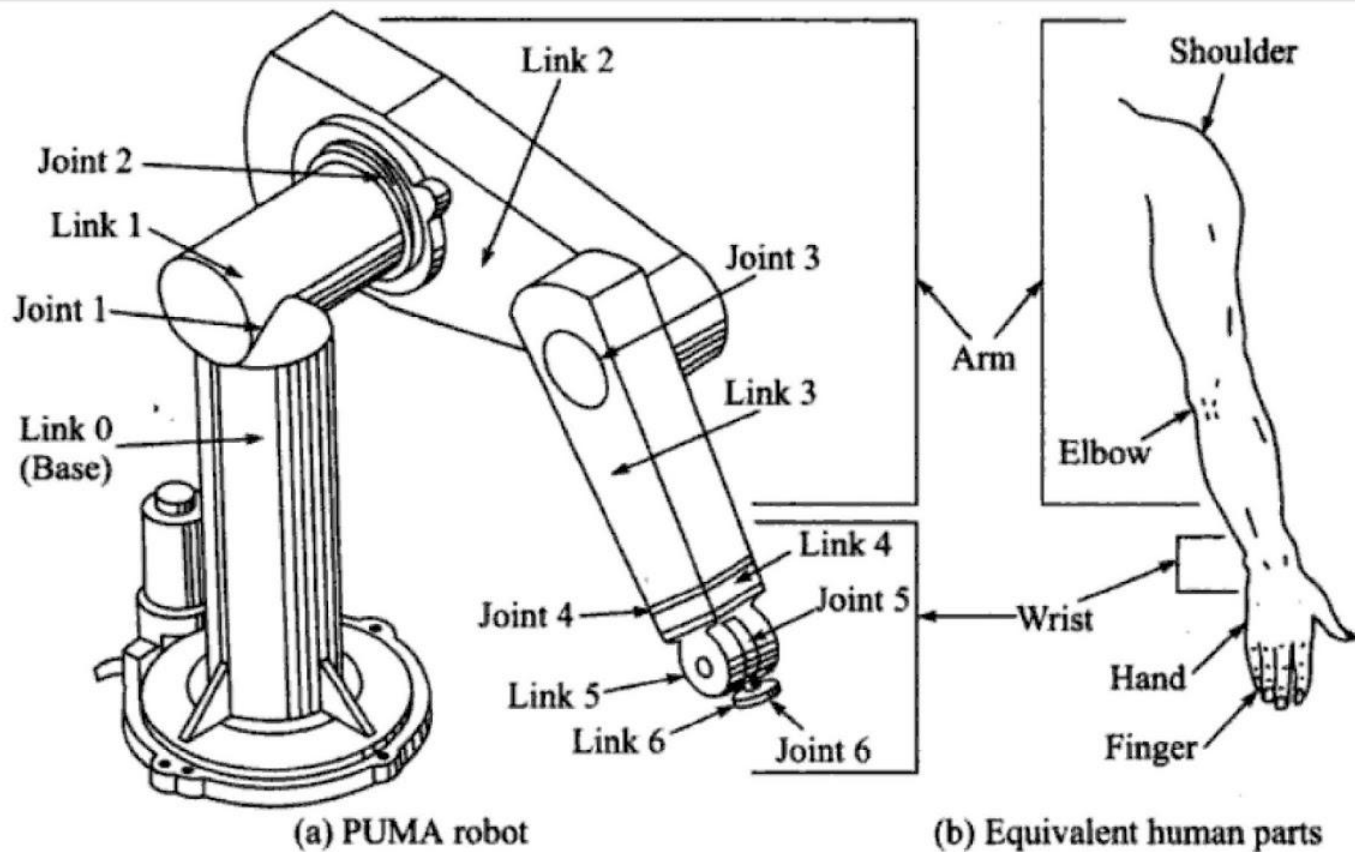
Subsystems







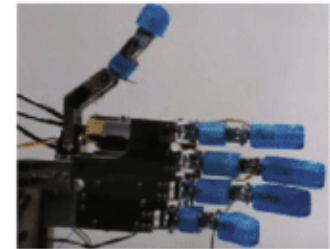
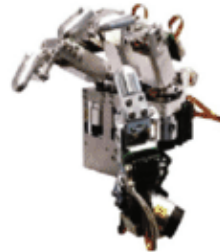
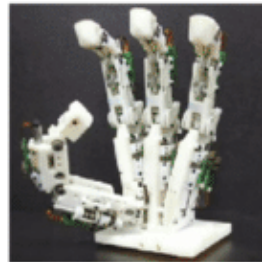
Manipulator



End Effector



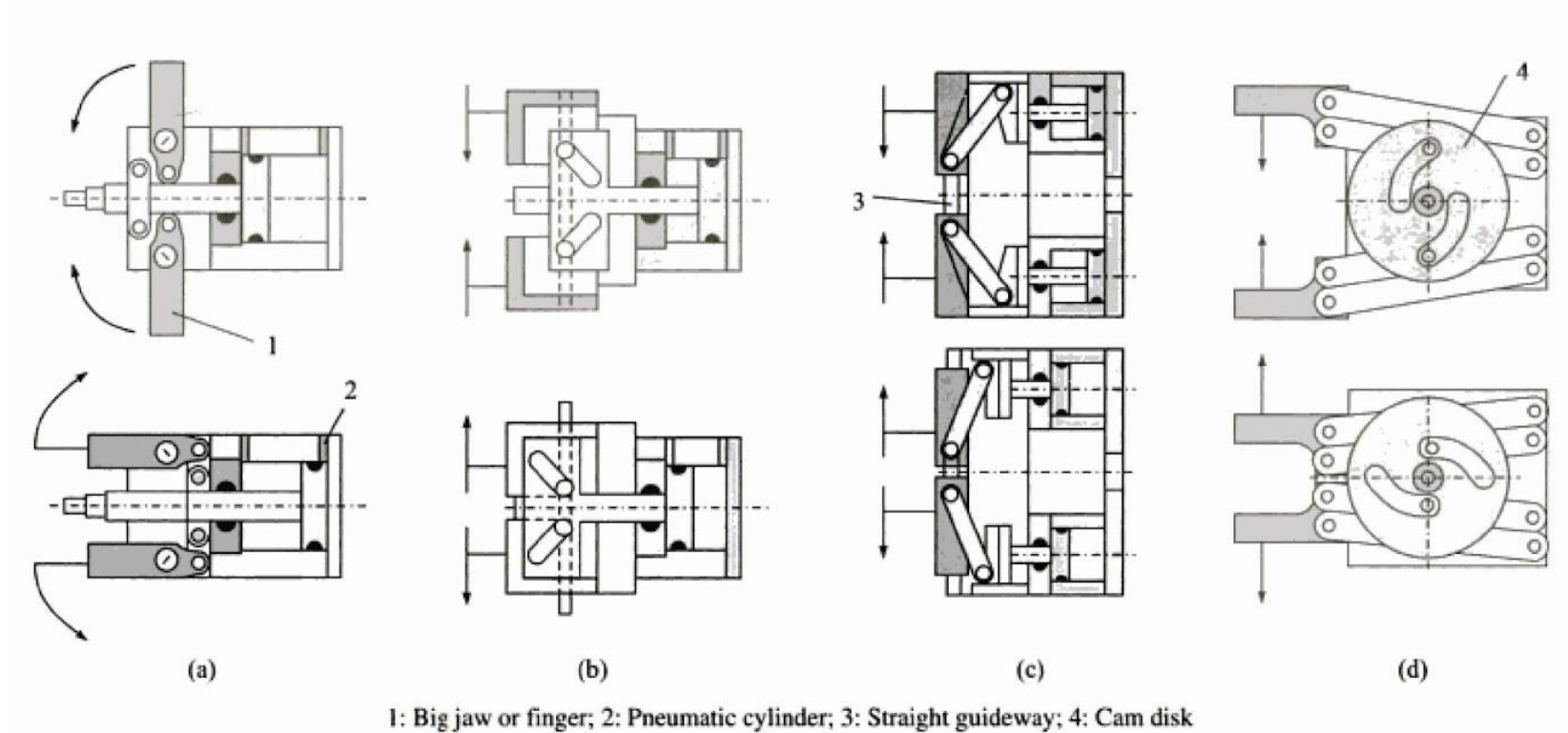
(a)



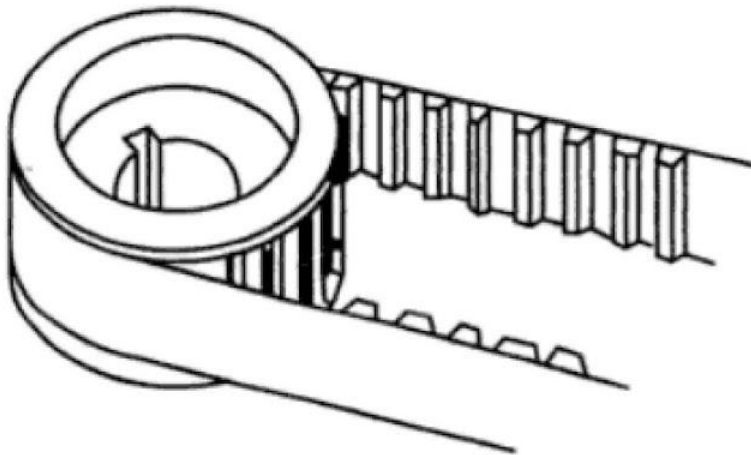
(b)

Two types of fingered end-effectors: (a) gripper type, (b) anthropomorphic type

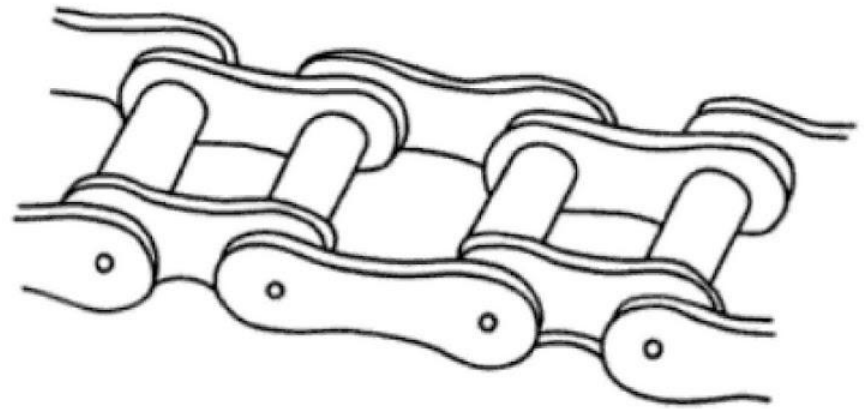
End-effector



Transmission (Belt and chain)



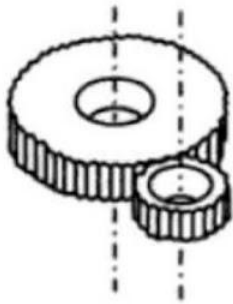
(a) Synchronous belt



(b) Roller chain

Categories	Synchronous Belt	Roller Chain
Efficiency	Generally more efficient, with lower friction losses	Less efficient due to higher friction and potential for more energy loss.
Noise	Quieter operation.	Noisier due to metal-on-metal contact.
Maintenance	Requires less maintenance, but belts can wear out and need replacement.	Requires regular lubrication and tension adjustment; chains are more durable but can rust and need more frequent maintenance.
Durability	More susceptible to stretching and breaking under high loads.	Better suited for high-load and high-torque applications due to its strength.
Precision	Less precise due to potential for slippage	More precise with minimal slippage, providing consistent motion
Cost	Typically cheaper to produce and replace	More expensive initially and over time due to maintenance needs
Flexibility	More flexible, can handle more complex routing	Less flexible, limited to simpler routing paths
Speed Range	Better for high-speed applications	Better for lower speed, high-torque applications
Weight	Lighter weight	Heavier due to metal construction
Environmental Resistance	More susceptible to temperature changes and can degrade with exposure to certain chemicals	More robust in harsh environments, better resistance to temperature extremes and chemicals
Elasticity	Elastic, which can dampen vibrations but may cause delay in response	Inelastic, providing more immediate response to changes in load

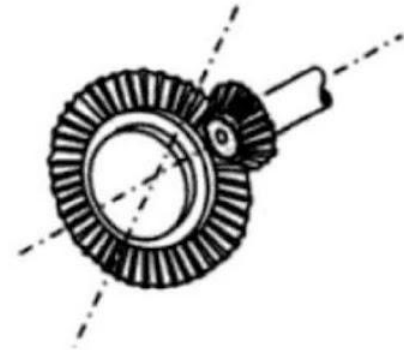
Transmission (Gears)



Spur gears



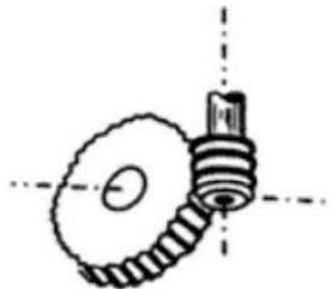
Helical gears



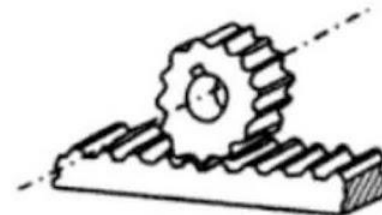
Straight bevel



Spiral bevel



Worm



Rack and pinion

1. Spur Gear -

- i. Parallel shaft gear, cylindrical, rotary motion
- ii. Tooth straight and parallel to shaft
- iii. High accuracy, easy production, no load, slow

2. Helical Gear -

- i. Parallel shaft gear, cylindrical
- ii. Winding tooth lines, better meshing
- iii. Heavy lifting, high speed

3. Straight Bevel -

- i. Cone shaped
- ii. Power transmitted at 90 degrees angle

4. Spiral Bevel -

- i. Cone shaped, teeth are curved
- ii. More efficient, strong, quiet
- iii. Harder Production

5. Worm -

- i. Screw shape cut on shaft (worm), mating gear (worm wheel)
- ii. Can be of various shapes, power to different axis
- iii. Low efficiency, smooth, quiet, high torque

6. Rack and Pinion -

- i. Rotational to linear motion
- ii. High precision
- iii. High load capacity



Spur Gear



Helical Gear



Bevel Gear



Spiral Bevel Gear



Worm Gear



Rack and Pinion

Actuators



Pneumatic Actuator

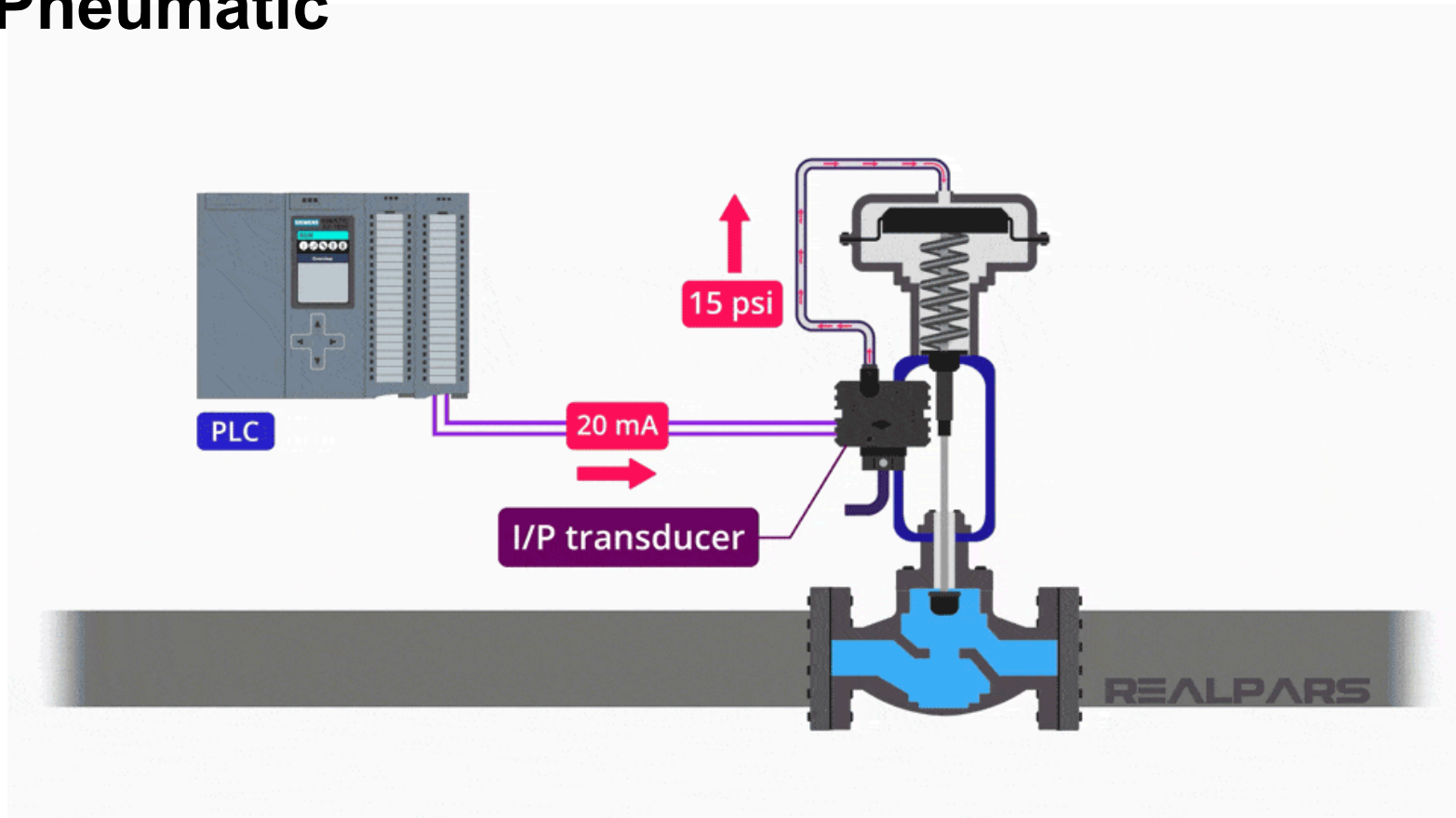


Hydraulic Actuator



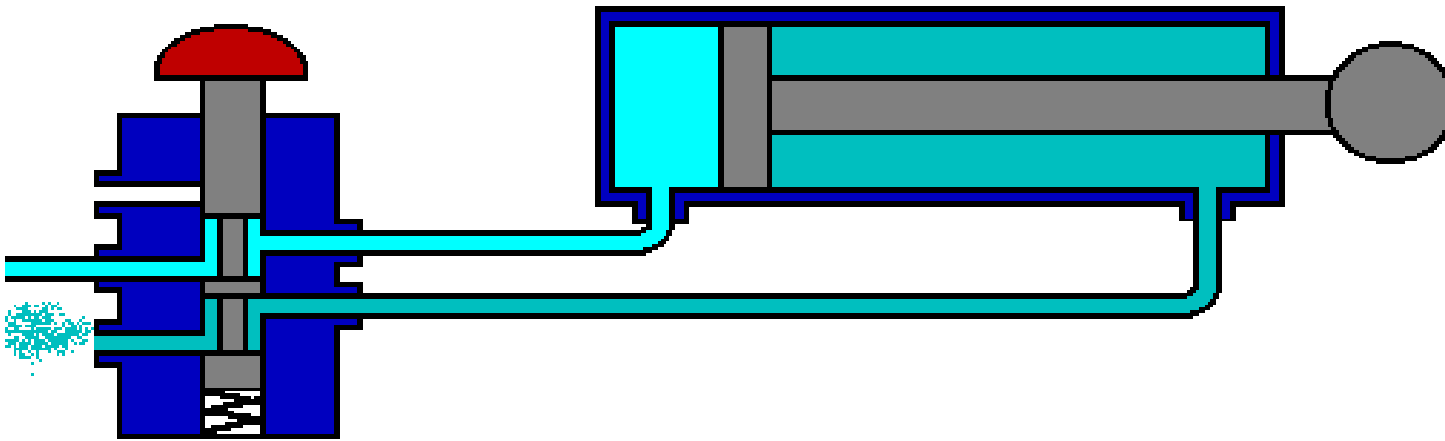
Electric Actuator

Pneumatic



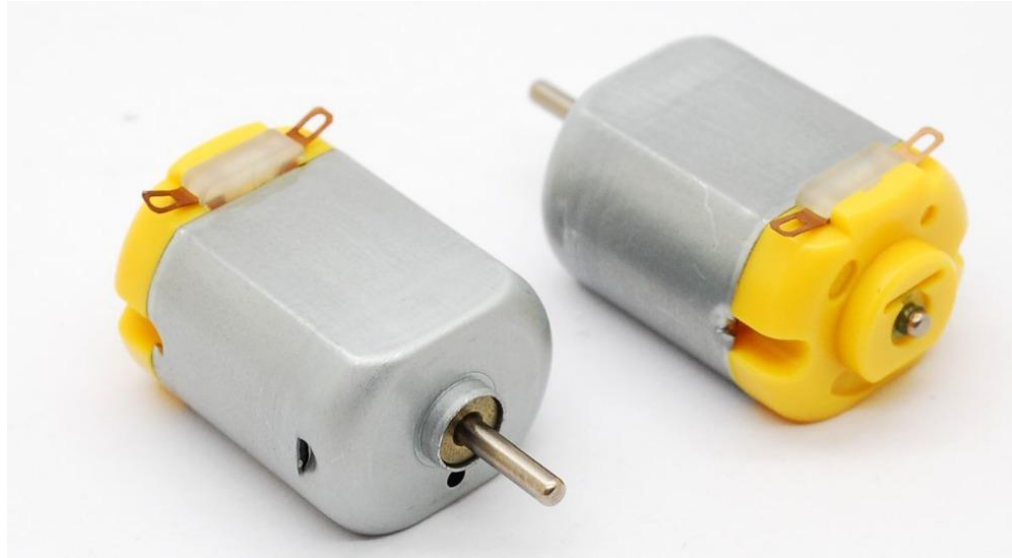
Pneumatic actuators use **compressed air** to produce linear or rotary motion. They are **fast, clean, and suitable for light to medium** load applications. Commonly used in automation where **speed and simplicity** are important.

Hydraulic



Hydraulic actuators operate using **pressurized fluid** to generate **high force and torque**. They are ideal for **heavy-duty applications** requiring **precise and powerful motion**. These actuators are widely used in **construction and industrial machinery**.

Electric

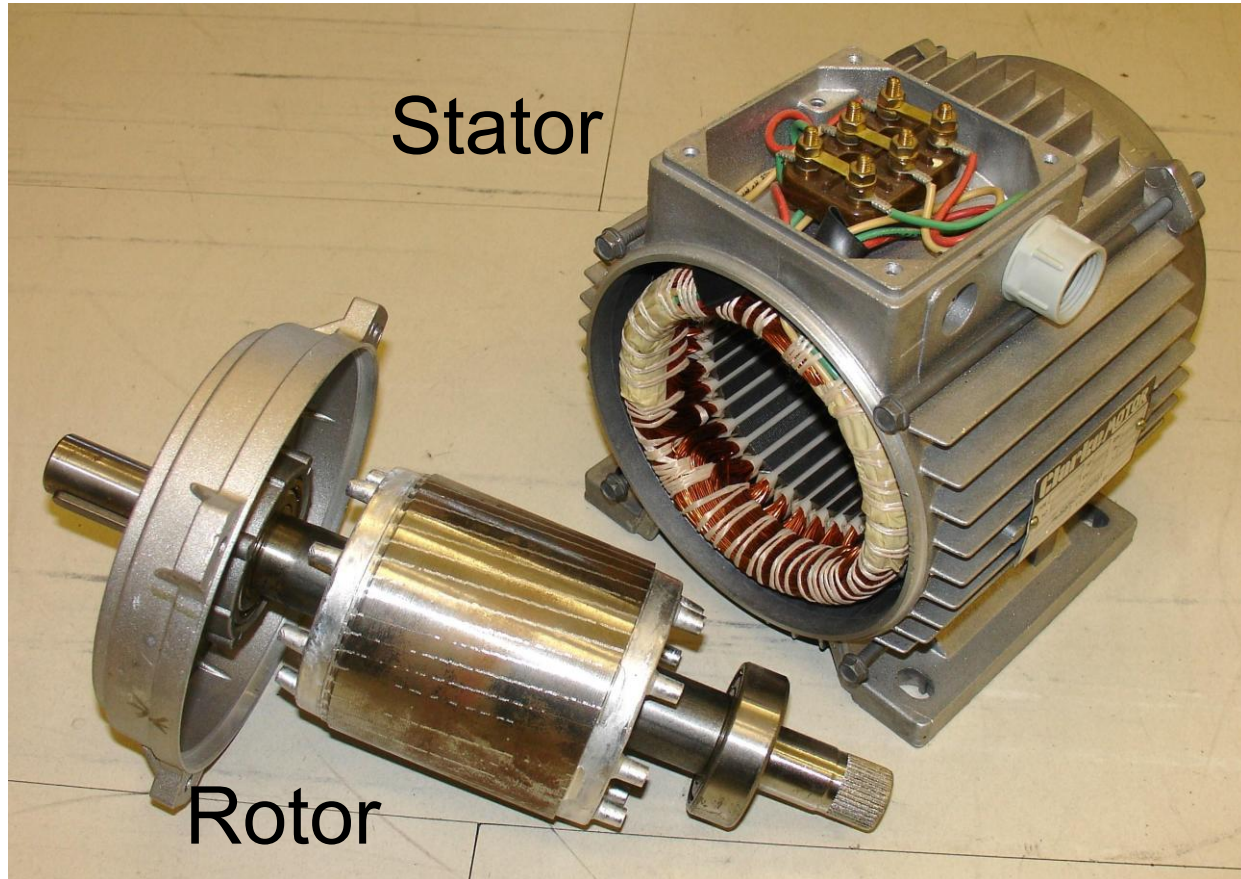


Electric actuators convert **electrical energy into mechanical motion** using motors. They offer **high precision, easy control, and low maintenance** compared to fluid-based systems. Suitable for applications requiring **accurate positioning** and integration with digital control systems.

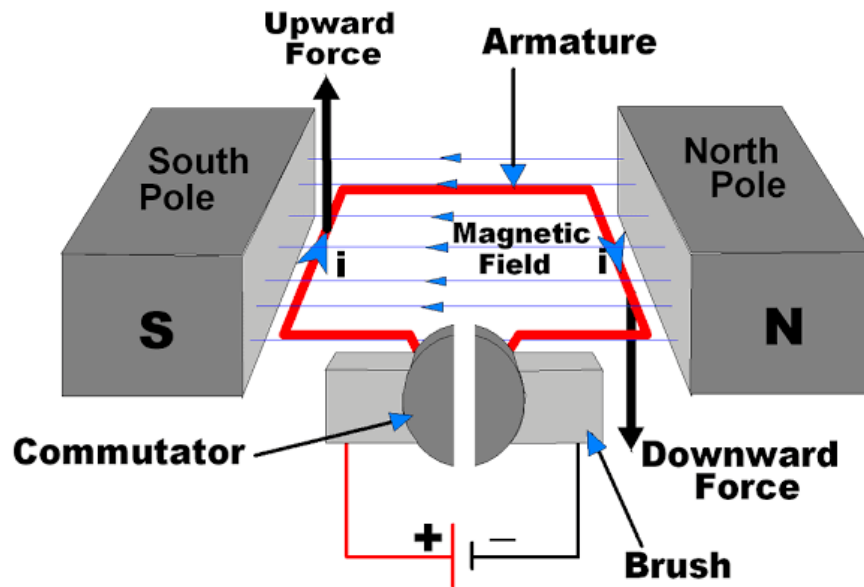
Actuators Comparison

Characteristics	Pneumatic	Hydraulic	Electric
Efficiency	Low	Low	High
Reliability	Excellent	Good	Good
Maintenance	High user-maintenance	High user-maintenance	Little to no maintenance
Purchase Cost	Low	High	High
Operating Cost	Moderate	High	Low

Motor



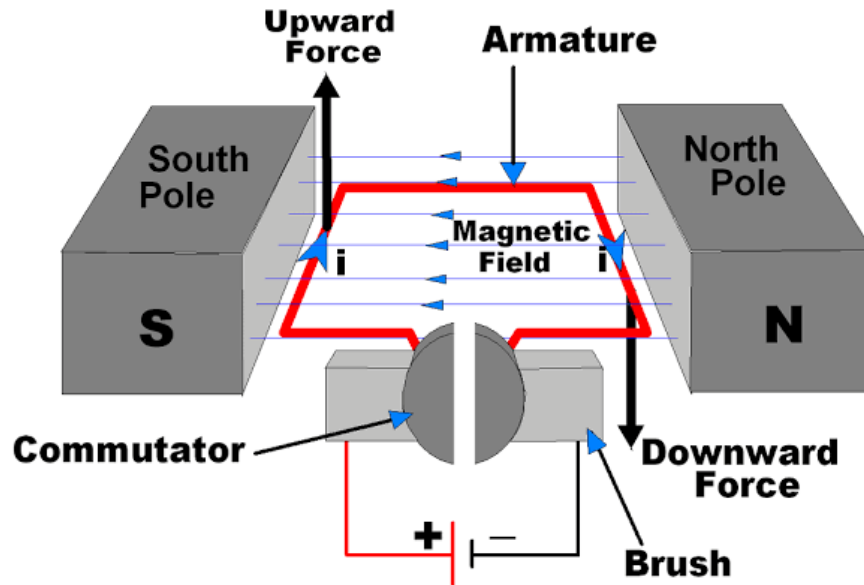
DC motor



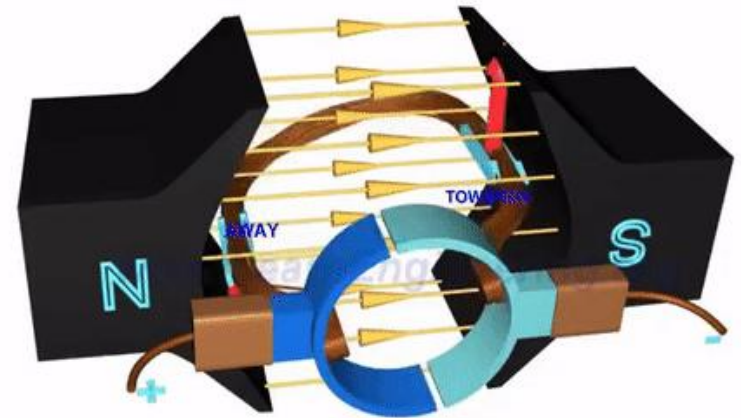
DC Motor Conceptual Diagram

Working mechanism: <https://youtu.be/LAtPHANefQo>

DC motor

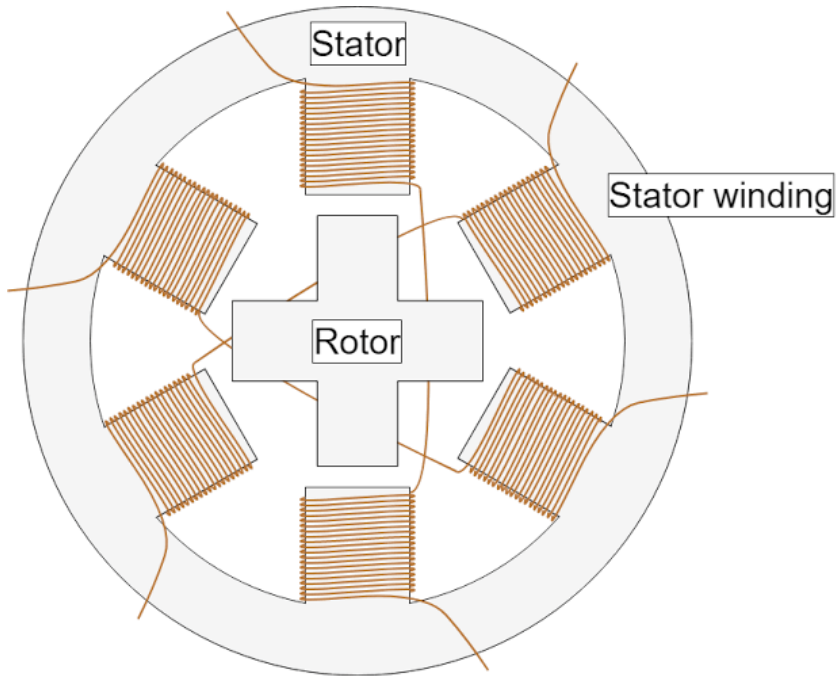


DC Motor Conceptual Diagram



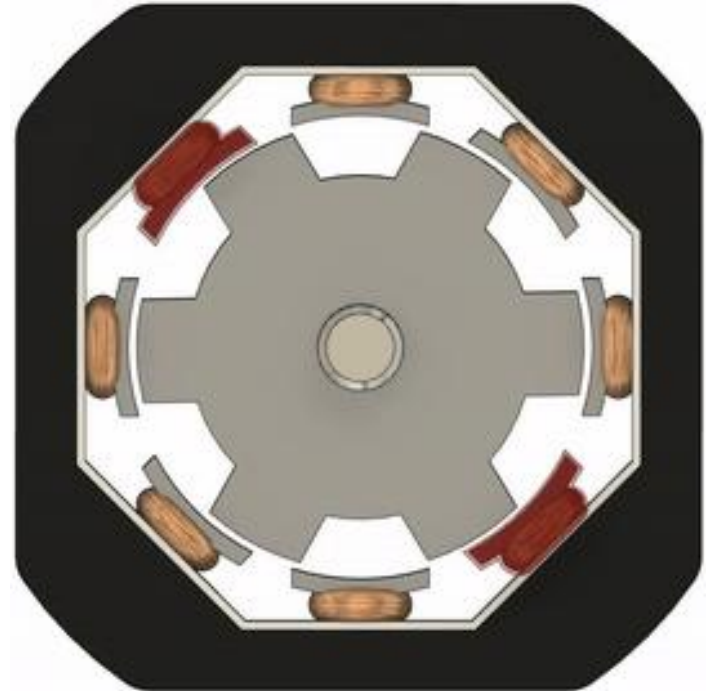
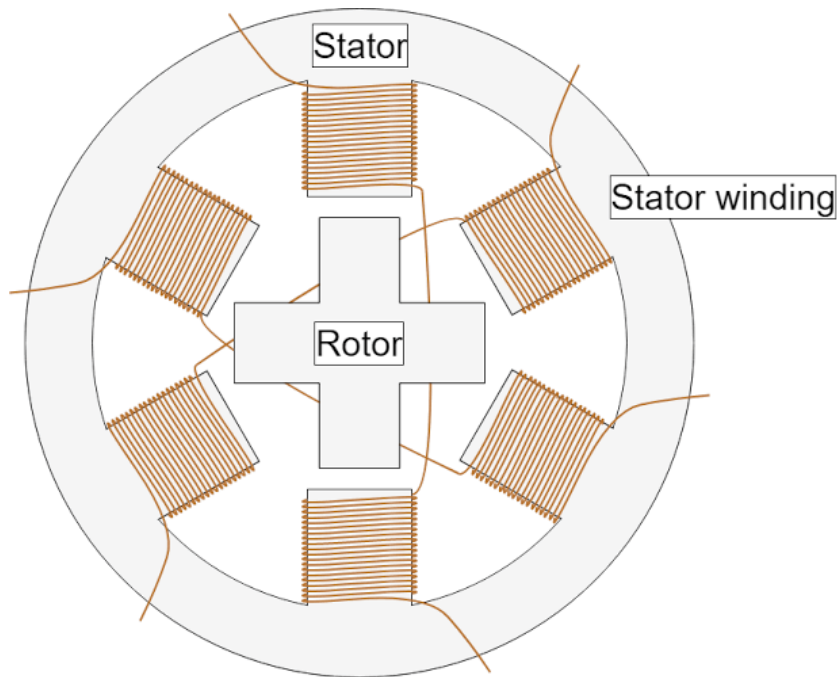
A DC motor works by converting electrical energy into **mechanical rotation through electromagnetic interaction**. When current flows through the armature winding, it creates a magnetic field that interacts with the stator field, producing torque. A commutator is used to maintain continuous rotation by reversing current direction. Speed control is typically achieved by varying the supply voltage.

Stepper Motor



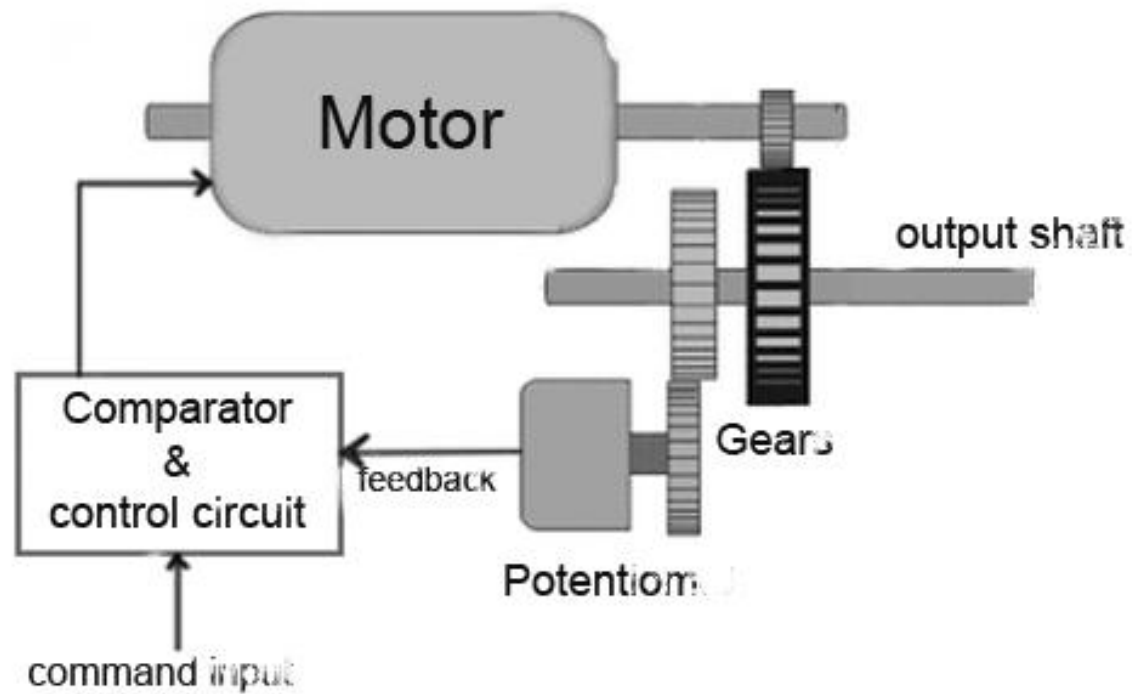
Working mechanism: <https://youtu.be/eyqwLiowZiU>

Stepper Motor

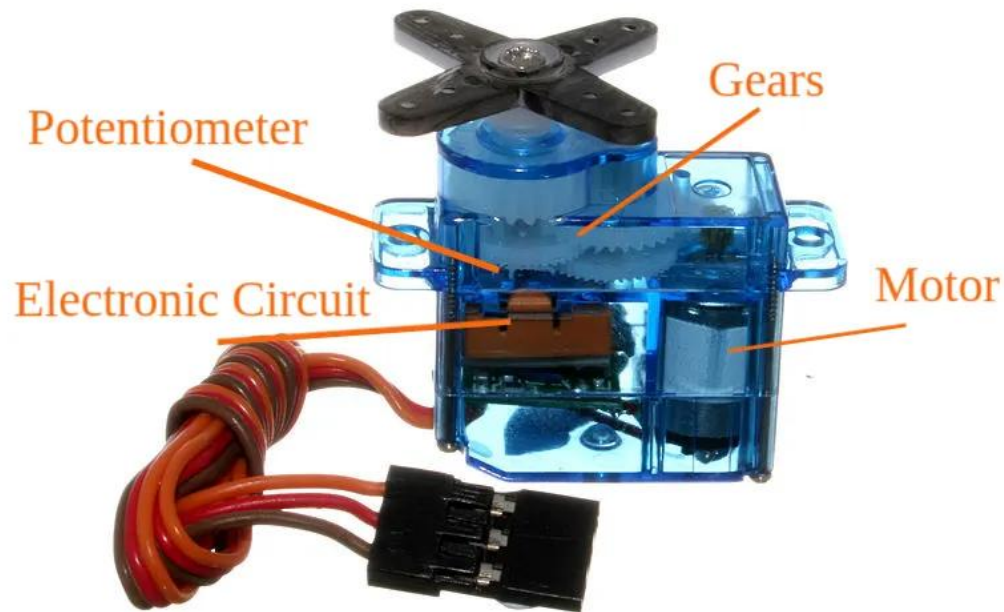


A stepper motor operates by dividing a full rotation into a **series of equal, discrete steps**. Sequential energizing of stator windings creates a rotating magnetic field that moves the rotor step by step. Each input pulse corresponds to a fixed angular movement, enabling precise position control. Stepper motors can hold their position without feedback when energized.

Servo Motor

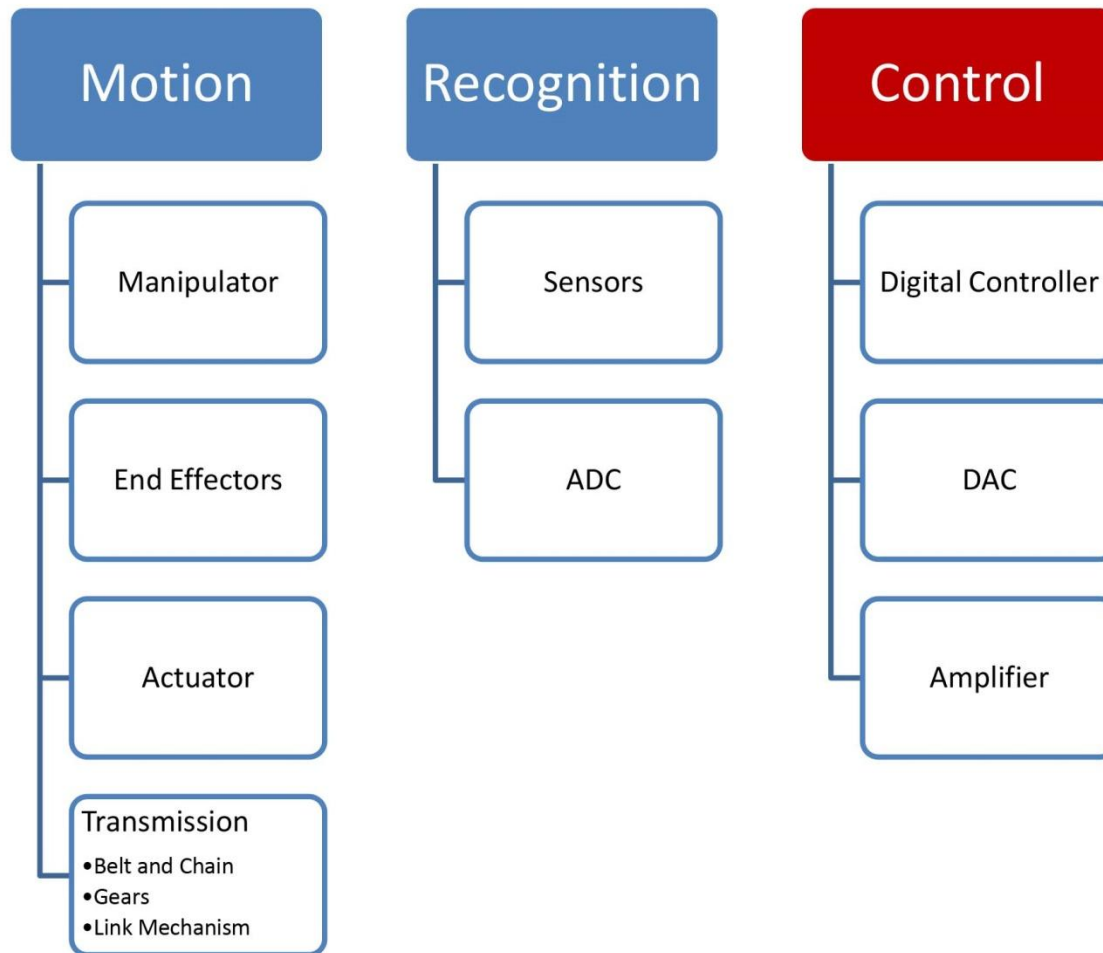


Servo Motor



A servo motor works using a **closed-loop control system with feedback** from a potentiometer. The controller compares the desired position with the actual position and adjusts the motor accordingly. This allows high precision and accuracy in position, speed, and torque control. Servo motors are widely used in robotics and automation where precision is critical.

Subsystems

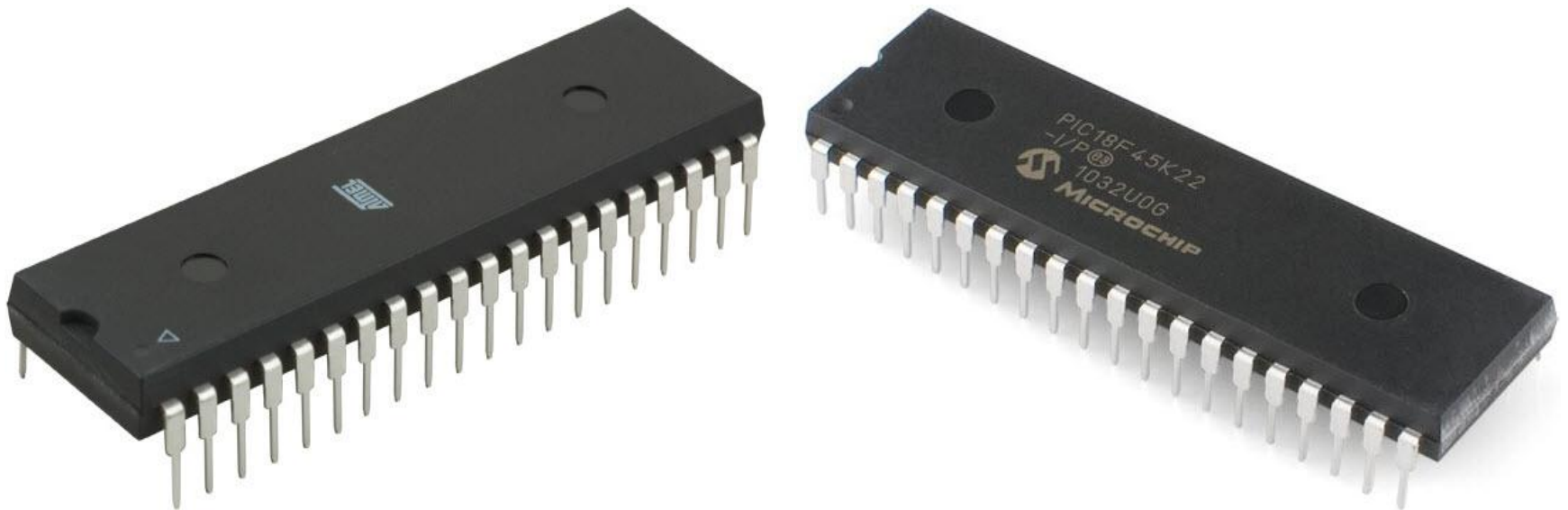


Control Hardware

- Microcontrollers
- Arduino
- PLC
- FPGA
- Single Board Computer
- Portable PCs
- Cloud Computing System

Microcontrollers

A microcontroller is a small, integrated circuit that contains a processor, memory, and input/output peripherals on a single chip. In robotics, it is used for low-level control tasks such as sensor reading, motor control, and real-time decision making in embedded systems.



Arduino

Arduino is an open-source microcontroller platform designed for rapid prototyping and ease of use. In robotics, it is widely used for educational robots and simple control tasks like line following, obstacle detection, and actuator control



Arduino



Arduino Uno



Arduino Leonardo



Arduino Mega 2560



Arduino LilyPad



Arduino Mega ADK



Arduino Fio



Arduino Ethernet



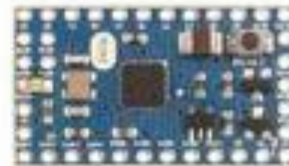
Arduino Pro



Arduino BT



Arduino Nano

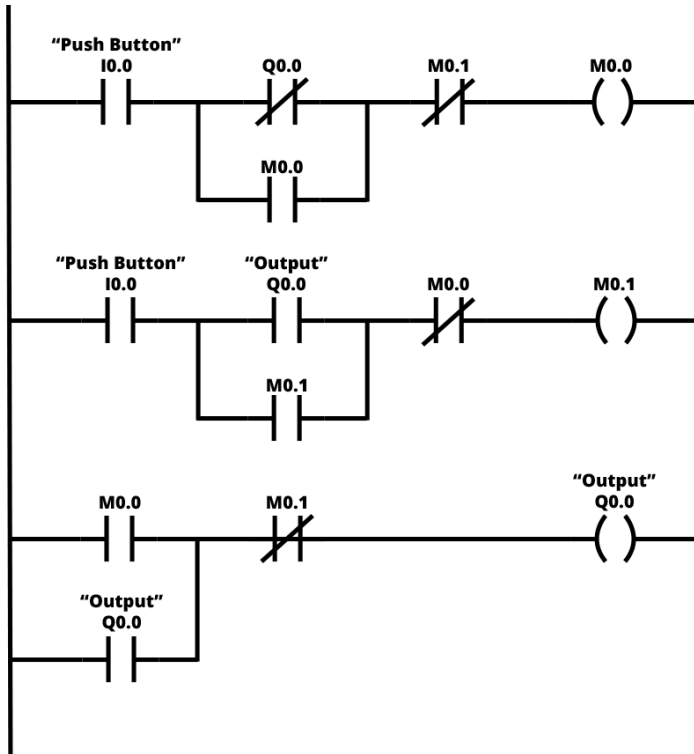


Arduino Mini



Arduino Pro Mini

Programmable Logic Controller (PLC)

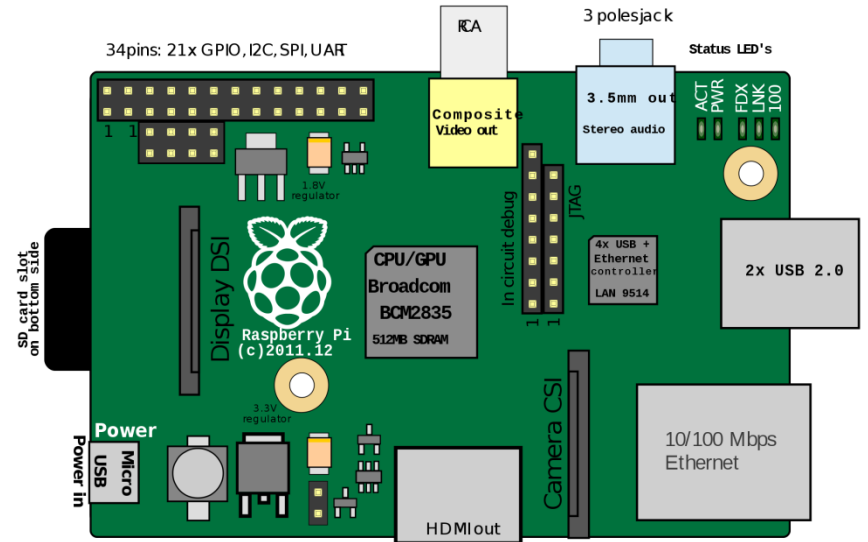


A PLC is an industrial-grade controller designed for reliable and deterministic control. In robotics, PLCs are used in industrial automation, robotic arms, and assembly lines where robustness and real-time performance are critical

Programmable Logic Controller (PLC)



Single board Computer



An SBC is a complete computer built on a single board with an operating system. In robotics, SBCs are used for high-level tasks such as computer vision, AI inference, SLAM, and communication, often working alongside microcontrollers for low-level control.

Raspberry Pi 4, Tinkerboard



Raspberry Pi: Use when you need **general-purpose computing** and moderate robotics tasks like control, networking, and basic vision.

ASUS Tinker Board: Use when **higher CPU performance** and better multimedia handling than Raspberry Pi are needed.

Little Panda and Jetson nano



Jetson Nano: Use when robotics applications require **GPU acceleration** for AI, deep learning, and real-time computer vision.

Littlepanda: Use when **compact size and basic SBC functionality** are sufficient for simple embedded or educational robotics projects.

Feature	Microcontroller	Microprocessor
Integration	Includes CPU, memory (RAM/ROM), and peripherals on a single chip	Consists mainly of CPU; external memory and peripherals are required
Cost	Low cost due to high integration	Higher cost because of additional external components
Power Consumption	Low power consumption, suitable for embedded systems	Higher power consumption
Application	Used in dedicated control-oriented applications (e.g., appliances, IoT)	Used in general-purpose computing (e.g., PCs, laptops)
Complexity	Simple hardware and software design	More complex system design
Real-time Control	Well suited for real-time and deterministic tasks	Less suitable for real-time control without additional hardware

Thank You